

Muhammad Hammad Bashir

Computer Vision Engineer — Sports AI —
Multimodal AI Systems

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Professional Summary

Computer Vision Engineer with 5+ years of experience building real-time AI systems across sports analytics, neuromarketing, agriculture, and multimodal AI. Specialized in end-to-end video intelligence systems involving object detection, tracking, pose estimation, and event recognition using YOLO, MediaPipe, ByteTrack, and deep learning pipelines. Strong focus on low-latency production systems, multi-camera streaming analytics, and scalable MLOps deployments.

Professional Experience

June 2025 – **Computer Vision Engineer (Sports AI)**, Sportunity / Sports Analytics Platform

- Present ○ Built a real-time sports computer vision system for soccer, tennis, and volleyball using live RTSP (HIKvision) camera feeds.
- Implemented YOLOv8m + ByteTrack + TrackNet for player, ball detection, and multi-object tracking under low-latency constraints.
- Developed event detection pipelines including ball bounce detection, trajectory interpolation, speed/acceleration estimation, and shot classification.
- Designed multi-camera synchronization system for cross-angle tracking, enabling accurate match reconstruction and highlight generation.
- Integrated MediaPipe pose estimation for biomechanics analysis (jump, sprint, movement efficiency).
- Optimized inference pipeline using ONNX + TensorRT for edge and GPU deployment.
- Built automated AI coaching system generating real-time analytics dashboards for coaches and athletes.

June 2023 – **AI Consultant (Computer Vision & Generative AI)**, Markify / CreativeScore AI, London, UK

- Developed neuromarketing AI systems analyzing ad effectiveness using computer vision and multi-modal signals (image, video, audio, and text).
- Built attention prediction and saliency mapping models to identify high-impact visual regions in advertisements.
- Designed emotion recognition pipelines for viewer reaction analysis using deep CNNs and transformer-based encoders.
- Implemented Pix2PixHD, GAN-based augmentation, and diffusion-based creative optimization for ad generation.
- Built end-to-end FastAPI + AWS + Azure CI/CD pipelines for scalable deployment.
- Delivered creative intelligence platform (creativescore.ai) for optimizing marketing performance and engagement.

Sept 2021 – **Research Assistant (AI & Computer Vision)**, LUMS / AI & Big Data Centre, Pakistan

- June 2023 ○ Developed ML-based livestock health monitoring systems using image classification and sensor data.
- Worked on flood and drought prediction using LSTM and RNN-based time-series models.
- Built agricultural AI systems for crop monitoring and yield prediction using satellite imagery.
- Implemented classical ML pipelines using Scikit-learn, XGBoost, Flask, and SQL-based deployments.

Aug 2021 – **Freelance ML Engineer**, Upwork & Fiverr

- Jul 2023 ○ Delivered end-to-end ML and computer vision solutions for global clients in sports analytics, document AI, and geospatial intelligence.

Selected Projects (High Impact)

- **Real-Time Multi-Camera Sports Vision System** Built a production-grade sports AI system using live RTSP feeds for real-time match analysis. Integrated YOLOv8, ByteTrack, and TrackNet for detection and tracking of players and ball. Developed ball interpolation, bounce detection, and event recognition modules. Used MediaPipe for pose-based biomechanics. Enabled multi-camera synchronization for highlight generation and AI-based coaching feedback.
- **MoversAI – Intelligent Inventory Quote Generation System** Developed an AI-powered web platform that allows users to capture room-by-room images for automated moving quotes. Used object detection and scene understanding to identify items, estimate volume, and generate pricing. Integrated human-in-the-loop verification for accuracy and production reliability.
- **Drone & Satellite Vision Intelligence (TMGDronity)** Worked on multimodal geospatial AI system combining satellite, drone, RGB, multispectral, and hyperspectral imagery. Implemented geospatial alignment using QGIS, GDAL, and rasterio. Built YOLO-based object detection and UNet segmentation models for vegetation, infrastructure, and underground object inference. Included greenery estimation and plant counting pipelines.
- **DoctorAI – Fall Detection & Health Monitoring System** Developed AI-assisted healthcare monitoring system combining computer vision and LLM-based reasoning. Implemented fall detection using pose estimation and anomaly detection pipelines. Integrated multimodal signals for patient activity monitoring and alert generation.

Skills

- **Computer Vision:** YOLOv8, OpenCV, MediaPipe, ByteTrack, DeepSORT, TrackNet, SAM, CLIP
- **Video Analytics:** RTSP pipelines, multi-camera sync, event detection, tracking systems
- **Deep Learning:** PyTorch, TensorFlow, CNNs, Transformers, GANs, diffusion models
- **MLOps:** Docker, CI/CD, Azure DevOps, MLflow, DVC
- **Deployment:** FastAPI, AWS, Azure, ONNX, TensorRT, serverless GPU systems
- **Geospatial AI:** QGIS, GDAL, rasterio, satellite image analysis
- **LLMs & Multimodal AI:** GPT, Claude, LLaMA, LangChain, CrewAI